DO NOW — RECALL

Floating-point arithmetic and precision

Name: _____ Class: _____ Date: _____

1. Normalise positive $0.0100_2 \times 2^4$.

2. What is −1 in 8-bit two's complement?

3. Calculate 0.5×2^4.

4. Calculate binary $0.1000_2 + 0.0100_2$.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

$0.1000_2 \times 2^3$.

2. Model answer

11111111.

3. Model answer

8.

4. Model answer

0.1100_2 = decimal 0.75.

Vocabulary — know these terms

exponent alignment — Express operands using the same power of two.

rounding — Choose a nearby representable value for an exact result.

truncation — Discard bits beyond the available precision.

overflow — Numerical result exceeds the representable range.

precision loss — Significant binary information cannot be stored.

NEW LEARNING

Read, annotate and explain

Align exponents

Before adding mantissas, both terms need the same power of two. Shift B right one place when raising its exponent by one.

Add the mantissas

$0.1000_2 + 0.0100_2 = 0.1100_2$; keep the common exponent 3.

Check in decimal

$0.1100_2 \times 2^3 = 0.75 \times 8 = 6$, matching 4 + 2.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Calculate $0.1100_2 \times 2^3$ minus $0.1000_2 \times 2^2$.

Align

B=0.1000×2^2 becomes 0.0100×2^3.

Subtract

$0.1100 - 0.0100 = 0.1000$.

Check

$0.1000_2 \times 2^3 = 4$; this matches decimal $6 - 2$.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Why must exponents be aligned before binary floating-point addition?

Question 2

Calculate and normalise $0.1010 \times 2^3 - 0.1000 \times 2^2$. Retain all shown mantissa bits during working.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain how exponent alignment can lose precision when the mantissa has a fixed number of bits.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

When increasing an exponent by 1 during alignment, the mantissa must shift right 1 place to preserve its value.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.